Structural identifiability of the velocity-jump model with v2=v3

In[ ]:=
```
(* Define equations*)
sys1 = {D[n1[x, t], t] + v1 * D[n1[x, t], x] +
     l1 * n1[x, t] - l2 * p21 * n2[x, t] - l3 * p31 * n3[x, t],
        D[n2[x, t], t] + v * D[n2[x, t], x] -
     l1 * p12 * n1[x, t] + l2 * n2[x, t] - l3 * (1 - p31) * n3[x, t],
        D[n3[x, t], t] + v * D[n3[x, t], x] -
     l1 * (1 - p12) * n1[x, t] - l2 * (1 - p21) * n2[x, t] + l3 * n3[x, t]};
```

In[ ]:=
```
(* Obtain parameters in the above expression *)
params = Quiet[Complement[Variables[sys1], Join[{n1[x, t], n2[x, t], n3[x, t ]},
     Select[Variables[sys1], #[[0]][[1]] === n1 || #[[0]][[1]] === n2 || #[[0]][[1]] === n3 &]]]]
```

Out[ ]=
```
{l1, l2, l3, p12, p21, p31, v, v1}
```

In[ ]:=
```
(* Define observation function - LHS is the parameter to get rid of first *)
obs = n1[x, t] → n[x, t] - n2[x, t] - n3[x, t];
```

In[ ]:=
```
sys2 = sys1 /. obs;
sys3 = sys2 /. D[obs, {x, 1}];
sys4 = FullSimplify[sys3 /. D[obs, {t, 1}]]
```

Out[ ]=
$$\Big\{l1\, n[x,t] - (l1 + l2\, p21)\, n2[x,t] - (l1 + l3\, p31)\, n3[x,t] + n^{(0,1)}[x,t] - n2^{(0,1)}[x,t] - n3^{(0,1)}[x,t] + v1\left(n^{(1,0)}[x,t] - n2^{(1,0)}[x,t] - n3^{(1,0)}[x,t]\right),\\ -l1\, p12\, n[x,t] + (l2 + l1\, p12)\, n2[x,t] + (l1\, p12 + l3\,(-1 + p31))\, n3[x,t] + n2^{(0,1)}[x,t] + v\, n2^{(1,0)}[x,t],\\ l2\,(-1 + p21)\, n2[x,t] + l1\,(-1 + p12)\,(n[x,t] - n2[x,t] - n3[x,t]) + l3\, n3[x,t] + n3^{(0,1)}[x,t] + v\, n3^{(1,0)}[x,t]\Big\}$$

In[ ]:=
```
summedsys = FullSimplify[sys1[[1]] + sys1[[2]] + sys1[[3]] /.
    Table[D[obs, {t, i[[1]]}, {x, i[[2]]}], {i, {{0, 0}, {1, 0}, {0, 1}}}]]
```

Out[ ]=
$$n^{(0,1)}[x,t] + v1\, n^{(1,0)}[x,t] + (v - v1)\left(n2^{(1,0)}[x,t] + n3^{(1,0)}[x,t]\right)$$

In[ ]:=
```
solsummedsys = Solve[0 == summedsys, D[n2[x, t], {x, 1}]][[1]]
```

Out[ ]=
$$\left\{n2^{(1,0)}[x,t] \to \frac{n^{(0,1)}[x,t] + v1\, n^{(1,0)}[x,t] + v\, n3^{(1,0)}[x,t] - v1\, n3^{(1,0)}[x,t]}{-v + v1}\right\}$$

In[ ]:=
```
sys5 = D[sys4, {x, 1}] /. solsummedsys;
```

In[ ]:=
```
sys6 = sys5 /. D[solsummedsys, {x, 1}];
```

In[•]:=
```
sys7 = Flatten[sys6 /. D[solsummedsys, {t, 1}]]
```

Out[•]=

$$\Bigg\{l1\, n^{(1,0)}[x,t] - (l1 + l3\, p31)\, n3^{(1,0)}[x,t] - \frac{(l1 + l2\, p21)\left(n^{(0,1)}[x,t] + v1\, n^{(1,0)}[x,t] + v\, n3^{(1,0)}[x,t] - v1\, n3^{(1,0)}[x,t]\right)}{-v + v1} + n^{(1,1)}[x,t] - n3^{(1,1)}[x,t] - \frac{n^{(0,2)}[x,t] + v1\, n^{(1,1)}[x,t] + v\, n3^{(1,1)}[x,t] - v1\, n3^{(1,1)}[x,t]}{-v + v1} + v1\left(n^{(2,0)}[x,t] - n3^{(2,0)}[x,t] - \frac{n^{(1,1)}[x,t] + v1\, n^{(2,0)}[x,t] + v\, n3^{(2,0)}[x,t] - v1\, n3^{(2,0)}[x,t]}{-v + v1}\right),$$

$$-l1\, p12\, n^{(1,0)}[x,t] + (l1\, p12 + l3\,(-1 + p31))\, n3^{(1,0)}[x,t] + \frac{(l2 + l1\, p12)\left(n^{(0,1)}[x,t] + v1\, n^{(1,0)}[x,t] + v\, n3^{(1,0)}[x,t] - v1\, n3^{(1,0)}[x,t]\right)}{-v + v1} + \frac{n^{(0,2)}[x,t] + v1\, n^{(1,1)}[x,t] + v\, n3^{(1,1)}[x,t] - v1\, n3^{(1,1)}[x,t]}{-v + v1} + \frac{v\left(n^{(1,1)}[x,t] + v1\, n^{(2,0)}[x,t] + v\, n3^{(2,0)}[x,t] - v1\, n3^{(2,0)}[x,t]\right)}{-v + v1},$$

$$l3\, n3^{(1,0)}[x,t] + \frac{l2\,(-1 + p21)\left(n^{(0,1)}[x,t] + v1\, n^{(1,0)}[x,t] + v\, n3^{(1,0)}[x,t] - v1\, n3^{(1,0)}[x,t]\right)}{-v + v1} + l1\,(-1 + p12)\left(n^{(1,0)}[x,t] - n3^{(1,0)}[x,t] - \frac{n^{(0,1)}[x,t] + v1\, n^{(1,0)}[x,t] + v\, n3^{(1,0)}[x,t] - v1\, n3^{(1,0)}[x,t]}{-v + v1}\right) + n3^{(1,1)}[x,t] + v\, n3^{(2,0)}[x,t]\Bigg\}$$

In[•]:=
```
(* Define required order - the maximum needed should be 8 *)
requiredOrder = 4;
```

In[•]:=
```
(* The other variable(s) to eliminate *)
varElim = {n3}; varObs = n;
```

In[•]:=
```
(* All possible orderDifference derivatives to take *)
deriv = Flatten[Table[{i, j}, {i, 0, requiredOrder}, {j, 0, requiredOrder}], 1];
derivFull = Select[deriv, Total[#] ≤ requiredOrder &];
derivHigh = Select[deriv, Total[#] ≤ requiredOrder - 2 &];(* -1-1*)
derivLow = Select[deriv, Total[#] ≤ requiredOrder - 3 &];(* -2-1*)
```

In[•]:=
```
(* Unknown variables up to the required order *)
varUnknown = Flatten[
   Table[D[var[x, t], {x, d[[1]]}, {t, d[[2]]}], {d, derivFull}, {var, varElim}]];
```

In[•]:=
```
(* Expand each equation - only 2 and 3 cos the first one is -2-3 *)
sys8 = Flatten[Join[Table[D[sys7[[2]], {x, d[[1]]}, {t, d[[2]]}], {d, derivLow}],
    Table[D[sys7[[3]], {x, d[[1]]}, {t, d[[2]]}], {d, derivLow}]]];
```

In[•]:=
```
(* Create matrix system *)
M = Table[Coefficient[expr, var], {expr, sys8}, {var, varUnknown}];
b = FullSimplify[M.varUnknown - sys8];
```

In[•]:=
```
(* Determine number of polynomial expressions required *)
Mb = Join[M, ArrayReshape[b, {Length[M], 1}], 2]
numExpr = MatrixRank[Mb] - MatrixRank[M]
```

Out[ ]=

$$\left\{\left\{0,\,0,\,0,\,0,\,0,\,-\text{l3}+\text{l1 p12}+\text{l3 p31}+\frac{\text{l2 v}}{-\text{v}+\text{v1}}+\frac{\text{l1 p12 v}}{-\text{v}+\text{v1}}-\frac{\text{l2 v1}}{-\text{v}+\text{v1}}-\frac{\text{l1 p12 v1}}{-\text{v}+\text{v1}},\right.\right.$$
$$\frac{\text{v}}{-\text{v}+\text{v1}}-\frac{\text{v1}}{-\text{v}+\text{v1}},\,0,\,0,\,\frac{\text{v}^2}{-\text{v}+\text{v1}}-\frac{\text{v v1}}{-\text{v}+\text{v1}},\,0,\,0,\,0,$$
$$0,\,0,\,\frac{1}{\text{v}-\text{v1}}\left(\left(\text{l2}+\text{l1 p12}\right)\text{n}^{(0,1)}[\text{x},\,\text{t}]+\text{n}^{(0,2)}[\text{x},\,\text{t}]+\right.$$
$$\left.\left.\left(\text{l1 p12 v}+\text{l2 v1}\right)\text{n}^{(1,0)}[\text{x},\,\text{t}]+(\text{v}+\text{v1})\,\text{n}^{(1,1)}[\text{x},\,\text{t}]+\text{v v1}\,\text{n}^{(2,0)}[\text{x},\,\text{t}]\right)\right\},$$
$$\left\{0,\,0,\,0,\,0,\,0,\,0,\,-\text{l3}+\text{l1 p12}+\text{l3 p31}+\frac{\text{l2 v}}{-\text{v}+\text{v1}}+\frac{\text{l1 p12 v}}{-\text{v}+\text{v1}}-\frac{\text{l2 v1}}{-\text{v}+\text{v1}}-\frac{\text{l1 p12 v1}}{-\text{v}+\text{v1}},\right.$$
$$\frac{\text{v}}{-\text{v}+\text{v1}}-\frac{\text{v1}}{-\text{v}+\text{v1}},\,0,\,0,\,\frac{\text{v}^2}{-\text{v}+\text{v1}}-\frac{\text{v v1}}{-\text{v}+\text{v1}},\,0,\,0,\,0,$$
$$0,\,\frac{1}{\text{v}-\text{v1}}\left(\left(\text{l2}+\text{l1 p12}\right)\text{n}^{(0,2)}[\text{x},\,\text{t}]+\text{n}^{(0,3)}[\text{x},\,\text{t}]+\right.$$
$$\left.\left.\left(\text{l1 p12 v}+\text{l2 v1}\right)\text{n}^{(1,1)}[\text{x},\,\text{t}]+(\text{v}+\text{v1})\,\text{n}^{(1,2)}[\text{x},\,\text{t}]+\text{v v1}\,\text{n}^{(2,1)}[\text{x},\,\text{t}]\right)\right\},$$
$$\left\{0,\,0,\,0,\,0,\,0,\,0,\,0,\,0,\,0,\,-\text{l3}+\text{l1 p12}+\text{l3 p31}+\frac{\text{l2 v}}{-\text{v}+\text{v1}}+\frac{\text{l1 p12 v}}{-\text{v}+\text{v1}}-\frac{\text{l2 v1}}{-\text{v}+\text{v1}}-\frac{\text{l1 p12 v1}}{-\text{v}+\text{v1}},\right.$$
$$\frac{\text{v}}{-\text{v}+\text{v1}}-\frac{\text{v1}}{-\text{v}+\text{v1}},\,0,\,\frac{\text{v}^2}{-\text{v}+\text{v1}}-\frac{\text{v v1}}{-\text{v}+\text{v1}},\,0,\,0,\,\frac{1}{\text{v}-\text{v1}}$$
$$\left(\left(\text{l2}+\text{l1 p12}\right)\text{n}^{(1,1)}[\text{x},\,\text{t}]+\text{n}^{(1,2)}[\text{x},\,\text{t}]+\left(\text{l1 p12 v}+\text{l2 v1}\right)\text{n}^{(2,0)}[\text{x},\,\text{t}]+\right.$$
$$\left.\left.(\text{v}+\text{v1})\,\text{n}^{(2,1)}[\text{x},\,\text{t}]+\text{v v1}\,\text{n}^{(3,0)}[\text{x},\,\text{t}]\right)\right\},$$
$$\left\{0,\,0,\,0,\,0,\,0,\,\text{l1}+\text{l3}-\text{l1 p12}+\frac{\text{l1 v}}{-\text{v}+\text{v1}}-\frac{\text{l2 v}}{-\text{v}+\text{v1}}-\frac{\text{l1 p12 v}}{-\text{v}+\text{v1}}+\frac{\text{l2 p21 v}}{-\text{v}+\text{v1}}-\right.$$
$$\frac{\text{l1 v1}}{-\text{v}+\text{v1}}+\frac{\text{l2 v1}}{-\text{v}+\text{v1}}+\frac{\text{l1 p12 v1}}{-\text{v}+\text{v1}}-\frac{\text{l2 p21 v1}}{-\text{v}+\text{v1}},\,1,\,0,\,0,\,\text{v},\,0,\,0,\,0,\,0,$$
$$\left.\frac{\left(\text{l1}-\text{l1 p12}+\text{l2}\,(-1+\text{p21})\right)\text{n}^{(0,1)}[\text{x},\,\text{t}]+\left(\text{l1}\,(\text{v}-\text{p12 v})+\text{l2}\,(-1+\text{p21})\,\text{v1}\right)\text{n}^{(1,0)}[\text{x},\,\text{t}]}{\text{v}-\text{v1}}\right\},$$
$$\left\{0,\,0,\,0,\,0,\,0,\,0,\,\text{l1}+\text{l3}-\text{l1 p12}+\frac{\text{l1 v}}{-\text{v}+\text{v1}}-\frac{\text{l2 v}}{-\text{v}+\text{v1}}-\frac{\text{l1 p12 v}}{-\text{v}+\text{v1}}+\right.$$
$$\frac{\text{l2 p21 v}}{-\text{v}+\text{v1}}-\frac{\text{l1 v1}}{-\text{v}+\text{v1}}+\frac{\text{l2 v1}}{-\text{v}+\text{v1}}+\frac{\text{l1 p12 v1}}{-\text{v}+\text{v1}}-\frac{\text{l2 p21 v1}}{-\text{v}+\text{v1}},\,1,\,0,\,0,\,\text{v},\,0,\,0,\,0,\,0,$$
$$\left.\frac{\left(\text{l1}-\text{l1 p12}+\text{l2}\,(-1+\text{p21})\right)\text{n}^{(0,2)}[\text{x},\,\text{t}]+\left(\text{l1}\,(\text{v}-\text{p12 v})+\text{l2}\,(-1+\text{p21})\,\text{v1}\right)\text{n}^{(1,1)}[\text{x},\,\text{t}]}{\text{v}-\text{v1}}\right\},$$
$$\left\{0,\,0,\,0,\,0,\,0,\,0,\,0,\,0,\,0,\,\text{l1}+\text{l3}-\text{l1 p12}+\frac{\text{l1 v}}{-\text{v}+\text{v1}}-\frac{\text{l2 v}}{-\text{v}+\text{v1}}-\right.$$
$$\frac{\text{l1 p12 v}}{-\text{v}+\text{v1}}+\frac{\text{l2 p21 v}}{-\text{v}+\text{v1}}-\frac{\text{l1 v1}}{-\text{v}+\text{v1}}+\frac{\text{l2 v1}}{-\text{v}+\text{v1}}+\frac{\text{l1 p12 v1}}{-\text{v}+\text{v1}}-\frac{\text{l2 p21 v1}}{-\text{v}+\text{v1}},\,1,\,0,\,\text{v},\,0,\,0,$$
$$\left.\left.\frac{\left(\text{l1}-\text{l1 p12}+\text{l2}\,(-1+\text{p21})\right)\text{n}^{(1,1)}[\text{x},\,\text{t}]+\left(\text{l1}\,(\text{v}-\text{p12 v})+\text{l2}\,(-1+\text{p21})\,\text{v1}\right)\text{n}^{(2,0)}[\text{x},\,\text{t}]}{\text{v}-\text{v1}}\right\}\right\}$$

Out[ ]=

1

In[•]:=
```
(* Perform row reduction, extract polynomial expression *)
{walltime, op} = AbsoluteTiming[
   RowReduce[Join[M, IdentityMatrix[Length[M]], 2]][[ ;; , -Length[M] ;;]];
walltime
expr = FullSimplify[op[[-numExpr ;;, ;;]].b]
```

Out[•]=
```
0.410768
```

Out[•]=
$$\left\{-\frac{1}{(l2+l3-l2\,p21)\,(v-v1)}\right.$$
$$\left((l1\,l2\,(-1+p12\,p21)+l2\,l3\,p21\,(-1+p31)-l2\,l3\,p31+l1\,l3\,(-1+p31-p12\,p31))\right.$$
$$n^{(0,1)}[x,t]-(l1+l2+l3)\,n^{(0,2)}[x,t]-n^{(0,3)}[x,t]-l1\,l2\,v\,n^{(1,0)}[x,t]-$$
$$l1\,l3\,v\,n^{(1,0)}[x,t]+l1\,l2\,p12\,p21\,v\,n^{(1,0)}[x,t]+l1\,l3\,p31\,v\,n^{(1,0)}[x,t]-$$
$$l1\,l3\,p12\,p31\,v\,n^{(1,0)}[x,t]-l2\,l3\,p21\,v1\,n^{(1,0)}[x,t]-$$
$$l2\,l3\,p31\,v1\,n^{(1,0)}[x,t]+l2\,l3\,p21\,p31\,v1\,n^{(1,0)}[x,t]-2\,l1\,v\,n^{(1,1)}[x,t]-$$
$$l2\,v\,n^{(1,1)}[x,t]-l3\,v\,n^{(1,1)}[x,t]-l2\,v1\,n^{(1,1)}[x,t]-l3\,v1\,n^{(1,1)}[x,t]-$$
$$2\,v\,n^{(1,2)}[x,t]-v1\,n^{(1,2)}[x,t]-l1\,v^2\,n^{(2,0)}[x,t]-l2\,v\,v1\,n^{(2,0)}[x,t]-$$
$$\left.\left.l3\,v\,v1\,n^{(2,0)}[x,t]-v^2\,n^{(2,1)}[x,t]-2\,v\,v1\,n^{(2,1)}[x,t]-v^2\,v1\,n^{(3,0)}[x,t]\right)\right\}$$

In[•]:=
```
expr2 = FullSimplify[expr / (1/((l2 + l3 - l2 p21) (v - v1)))]
```

Out[•]=
$$\{(l1\,(l2+l3-l2\,p12\,p21+l3\,(-1+p12)\,p31)+l2\,l3\,(p21+p31-p21\,p31))\,n^{(0,1)}[x,t]+$$
$$(l1+l2+l3)\,n^{(0,2)}[x,t]+n^{(0,3)}[x,t]+l1\,l2\,v\,n^{(1,0)}[x,t]+$$
$$l1\,l3\,v\,n^{(1,0)}[x,t]-l1\,l2\,p12\,p21\,v\,n^{(1,0)}[x,t]-l1\,l3\,p31\,v\,n^{(1,0)}[x,t]+$$
$$l1\,l3\,p12\,p31\,v\,n^{(1,0)}[x,t]+l2\,l3\,p21\,v1\,n^{(1,0)}[x,t]+$$
$$l2\,l3\,p31\,v1\,n^{(1,0)}[x,t]-l2\,l3\,p21\,p31\,v1\,n^{(1,0)}[x,t]+2\,l1\,v\,n^{(1,1)}[x,t]+$$
$$l2\,v\,n^{(1,1)}[x,t]+l3\,v\,n^{(1,1)}[x,t]+l2\,v1\,n^{(1,1)}[x,t]+l3\,v1\,n^{(1,1)}[x,t]+$$
$$2\,v\,n^{(1,2)}[x,t]+v1\,n^{(1,2)}[x,t]+l1\,v^2\,n^{(2,0)}[x,t]+l2\,v\,v1\,n^{(2,0)}[x,t]+$$
$$l3\,v\,v1\,n^{(2,0)}[x,t]+v^2\,n^{(2,1)}[x,t]+2\,v\,v1\,n^{(2,1)}[x,t]+v^2\,v1\,n^{(3,0)}[x,t]\}$$

In[•]:=
```
(* Extract coefficients *)
varKnown = Join[{n[x, t]}, Quiet[Select[Variables[expr2], #[[0]][[1]] === n &]]]
coef1 = FullSimplify[Flatten[Values@CoefficientRules[expr2, varKnown]]]
```

Out[•]=
$$\{n[x,t],\,n^{(0,1)}[x,t],\,n^{(0,2)}[x,t],\,n^{(0,3)}[x,t],\,n^{(1,0)}[x,t],$$
$$n^{(1,1)}[x,t],\,n^{(1,2)}[x,t],\,n^{(2,0)}[x,t],\,n^{(2,1)}[x,t],\,n^{(3,0)}[x,t]\}$$

Out[•]=
$$\{l1\,(l2+l3-l2\,p12\,p21+l3\,(-1+p12)\,p31)+l2\,l3\,(p21+p31-p21\,p31),\,l1+l2+l3,$$
$$1,\,l1\,(l2+l3-l2\,p12\,p21+l3\,(-1+p12)\,p31)\,v+l2\,l3\,(p21+p31-p21\,p31)\,v1,$$
$$2\,l1\,v+(l2+l3)\,(v+v1),\,2\,v+v1,\,v\,(l1\,v+(l2+l3)\,v1),\,v\,(v+2\,v1),\,v^2\,v1\}$$

In[•]:=
```
(* Try another way: upper triangularise the matrix *)
{walltime2, op2} =
  AbsoluteTiming[LinearAlgebra`Private`UpperTriangularizeMatrix[
     Join[M, IdentityMatrix[Length[M]], 2]][[ ;; , -Length[M] ;;]]];
walltime2
```

Out[•]=
```
0.185653
```

In[•]:=
```
(* Compute expression *)
expr2 = FullSimplify[op2[[-numExpr ;;, ;;]].b]
```

Out[•]=

$$\left\{\frac{1}{v-v1}(l2+l3-l3\,p31)^2\,(l2\,p21-l3\,p31)^2\right.$$
$$\Big((l1\,l2\,(-1+p12\,p21)+l2\,l3\,p21\,(-1+p31)-l2\,l3\,p31+l1\,l3\,(-1+p31-p12\,p31))$$
$$n^{(0,1)}[x,t]-(l1+l2+l3)\,n^{(0,2)}[x,t]-n^{(0,3)}[x,t]-l1\,l2\,v\,n^{(1,0)}[x,t]-$$
$$l1\,l3\,v\,n^{(1,0)}[x,t]+l1\,l2\,p12\,p21\,v\,n^{(1,0)}[x,t]+l1\,l3\,p31\,v\,n^{(1,0)}[x,t]-$$
$$l1\,l3\,p12\,p31\,v\,n^{(1,0)}[x,t]-l2\,l3\,p21\,v1\,n^{(1,0)}[x,t]-$$
$$l2\,l3\,p31\,v1\,n^{(1,0)}[x,t]+l2\,l3\,p21\,p31\,v1\,n^{(1,0)}[x,t]-2\,l1\,v\,n^{(1,1)}[x,t]-$$
$$l2\,v\,n^{(1,1)}[x,t]-l3\,v\,n^{(1,1)}[x,t]-l2\,v1\,n^{(1,1)}[x,t]-l3\,v1\,n^{(1,1)}[x,t]-$$
$$2\,v\,n^{(1,2)}[x,t]-v1\,n^{(1,2)}[x,t]-l1\,v^2\,n^{(2,0)}[x,t]-l2\,v\,v1\,n^{(2,0)}[x,t]-$$
$$\left.l3\,v\,v1\,n^{(2,0)}[x,t]-v^2\,n^{(2,1)}[x,t]-2\,v\,v1\,n^{(2,1)}[x,t]-v^2\,v1\,n^{(3,0)}[x,t]\Big)\right\}$$

In[•]:=
```
(* Extract coefficients *)
coef12 = FullSimplify[Flatten[Values@CoefficientRules[expr2, varKnown]]]
```

Out[•]=

$$\left\{\frac{1}{v-v1}(l2+l3-l3\,p31)^2\,(l2\,p21-l3\,p31)^2\right.$$
$$(l1\,l2\,(-1+p12\,p21)+l2\,l3\,p21\,(-1+p31)-l2\,l3\,p31+l1\,l3\,(-1+p31-p12\,p31)),$$
$$\frac{(l1+l2+l3)\,(l2+l3-l3\,p31)^2\,(l2\,p21-l3\,p31)^2}{-v+v1},$$
$$\frac{(l2+l3-l3\,p31)^2\,(l2\,p21-l3\,p31)^2}{-v+v1},\ \frac{1}{v-v1}(l2+l3-l3\,p31)^2\,(l2\,p21-l3\,p31)^2$$
$$(l1\,(l2\,(-1+p12\,p21)+l3\,(-1+p31-p12\,p31))\,v+l2\,l3\,(p21\,(-1+p31)-p31)\,v1),$$
$$-\frac{(l2+l3-l3\,p31)^2\,(l2\,p21-l3\,p31)^2\,(2\,l1\,v+(l2+l3)\,(v+v1))}{v-v1},$$
$$-\frac{(l2+l3-l3\,p31)^2\,(l2\,p21-l3\,p31)^2\,(2\,v+v1)}{v-v1},$$
$$-\frac{(l2+l3-l3\,p31)^2\,(l2\,p21-l3\,p31)^2\,v\,(l1\,v+(l2+l3)\,v1)}{v-v1},$$
$$-\frac{(l2+l3-l3\,p31)^2\,(l2\,p21-l3\,p31)^2\,v\,(v+2\,v1)}{v-v1},$$
$$\left.-\frac{(l2+l3-l3\,p31)^2\,(l2\,p21-l3\,p31)^2\,v^2\,v1}{v-v1}\right\}$$

In[ ]:=
```
coef22 = FullSimplify[coef12 / coef12[[3]]]
```

Out[ ]=

$$\left\{l1\,(l2+l3-l2\,p12\,p21+l3\,(-1+p12)\,p31)+l2\,l3\,(p21+p31-p21\,p31),\ l1+l2+l3,\right.$$
$$1,\ l1\,(l2+l3-l2\,p12\,p21+l3\,(-1+p12)\,p31)\,v+l2\,l3\,(p21+p31-p21\,p31)\,v1,$$
$$\left.2\,l1\,v+(l2+l3)\,(v+v1),\ 2\,v+v1,\ v\,(l1\,v+(l2+l3)\,v1),\ v\,(v+2\,v1),\ v^2\,v1\right\}$$